FORMATIVE ASSESSMENT – 1 (FA-1)

MATHEMATICS PROJECT WORK

Topic: Geometrical Verification of $(a + b)^2 = a^2 + 2ab + b^2$

PRELIMINARY INFORMATION

Name of the Student: .....

Class & Section: **Class 9 ( )**

Roll Number: .....

Name of the Lesson: **Polynomials**

Name of the Guide Teacher: .....

AIM

To geometrically verify the algebraic identity $(a + b)^2 = a^2 + 2ab + b^2$ by constructing a square of side $(a + b)$ and calculating the sum of areas of its four component regions.

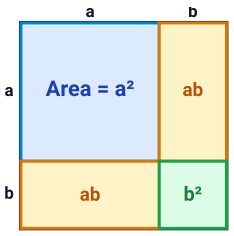
MATERIAL REQUIRED

- White Sheet / Drawing Sheet
- Ruler Scale
- Pencil & Eraser
- Colored Pencils / Glazed Paper
- Pen

PROCEDURE

1. **Construction:** Draw a large square ABCD of total side length $(a + b)$.
2. **Partitioning:** Divide the square into four parts: one square of side a (area = a^2), one square of side b (area = b^2), and two rectangles of dimensions $a \times b$ (area = ab each).
3. **Verification:** Substitute numerical values for a and b to verify that Total Area = $a^2 + 2ab + b^2$.

GEOMETRIC FIGURE (SQUARE OF SIDE A + B)



OBSERVATION AND VERIFICATION TABLE

S.No.	Values	Total Area: $(a + b)^2$	Sum of Parts: $a^2 + 2ab + b^2$	Observation
1	a = 4, b = 2	**$(4 + 2)^2 = 6^2 = 36$**	$4^2 + 2(4)(2) + 2^2 = 16 + 16 + 4 =$ **36**	**LHS = RHS (Verified)**
2	a = 3, b = 1	**$(3 + 1)^2 = 4^2 = 16$**	$3^2 + 2(3)(1) + 1^2 = 9 + 6 + 1 =$ **16**	**LHS = RHS (Verified)**
3	a = 5, b = 2	**$(5 + 2)^2 = 7^2 = 49$**	$5^2 + 2(5)(2) + 2^2 = 25 + 20 + 4 =$ **49**	**LHS = RHS (Verified)**

CONCLUSION

The total area of the square of side $(a + b)$ equals the sum of areas of its four component regions ($a^2 + ab + ab + b^2$). Hence, the identity $(a + b)^2 = a^2 + 2ab + b^2$ is verified geometrically.

Signature of the Student	Signature of the Guide Teacher
Date: _____	Date: _____